

Characteristics of the Radar Bright-band in Winter Storms

Olivia B. Samolej¹, Troy J. Zaremba², Robert M. Rauber¹

¹University of Illinois Urbana-Champaign, ²University of Washington



Introduction

In winter storms a transition between precipitation types often occurs that from south to north changes as ice particles fall from high in a cloud (Fig 1). In this study we are examining the perturbations in vertical radial velocity and reflectivity at W-band along and below the radar bright band in the transition zone between snow and rain, rain to freezing rain, freezing rain to sleet, and sleet to snow. The data are from the CRS radar on the NASA ER-2 aircraft and were collected during the NASA IMPACTS campaign.

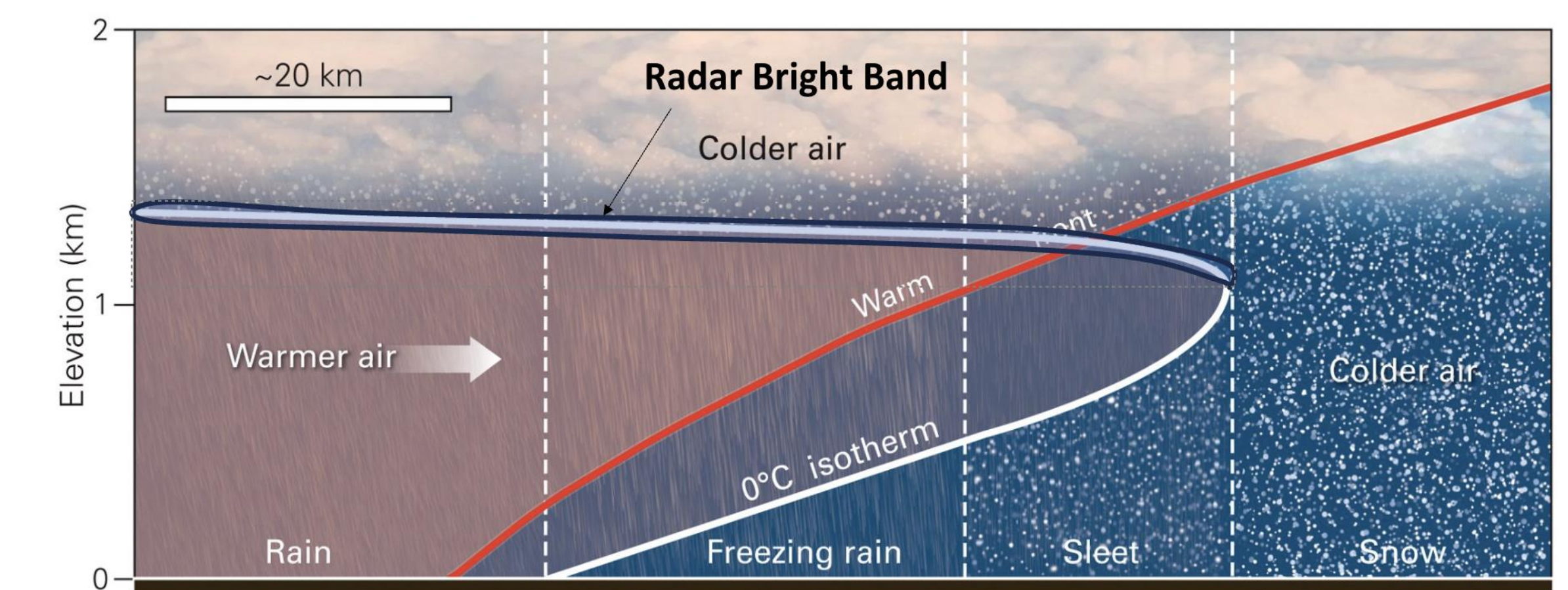


Fig. 1 Precipitation types across the comma head of a winter storm

IMPACTS

The NASA Investigation of Microphysics and Precipitation for Atlantic Coast-Threatening Snowstorms (IMPACTS) field campaign used the NASA P-3 Orion and ER-2 aircraft to collect in-situ microphysics and airborne radar data within the comma-head region of Atlantic Coast cyclones. It carried three radars: the ER-2 X-band Doppler Radar (EXRAD, 9.6 GHz) with conical/cross-track and nadir beams; the High-Altitude Imaging Wind and Rain Airborne Profiler (HIWRAP, Ku-13.9 GHz, Ka-35.6 GHz); and the nadir-pointing W-band Cloud Radar System (CRS, 94 GHz). This analysis uses CRS nadir-beam equivalent reflectivity factor (Z_e) and vertical radial velocity (V_r) to examine horizontal variability of cloud substructures and circulations, given its smaller beamwidth and higher sensitivity.

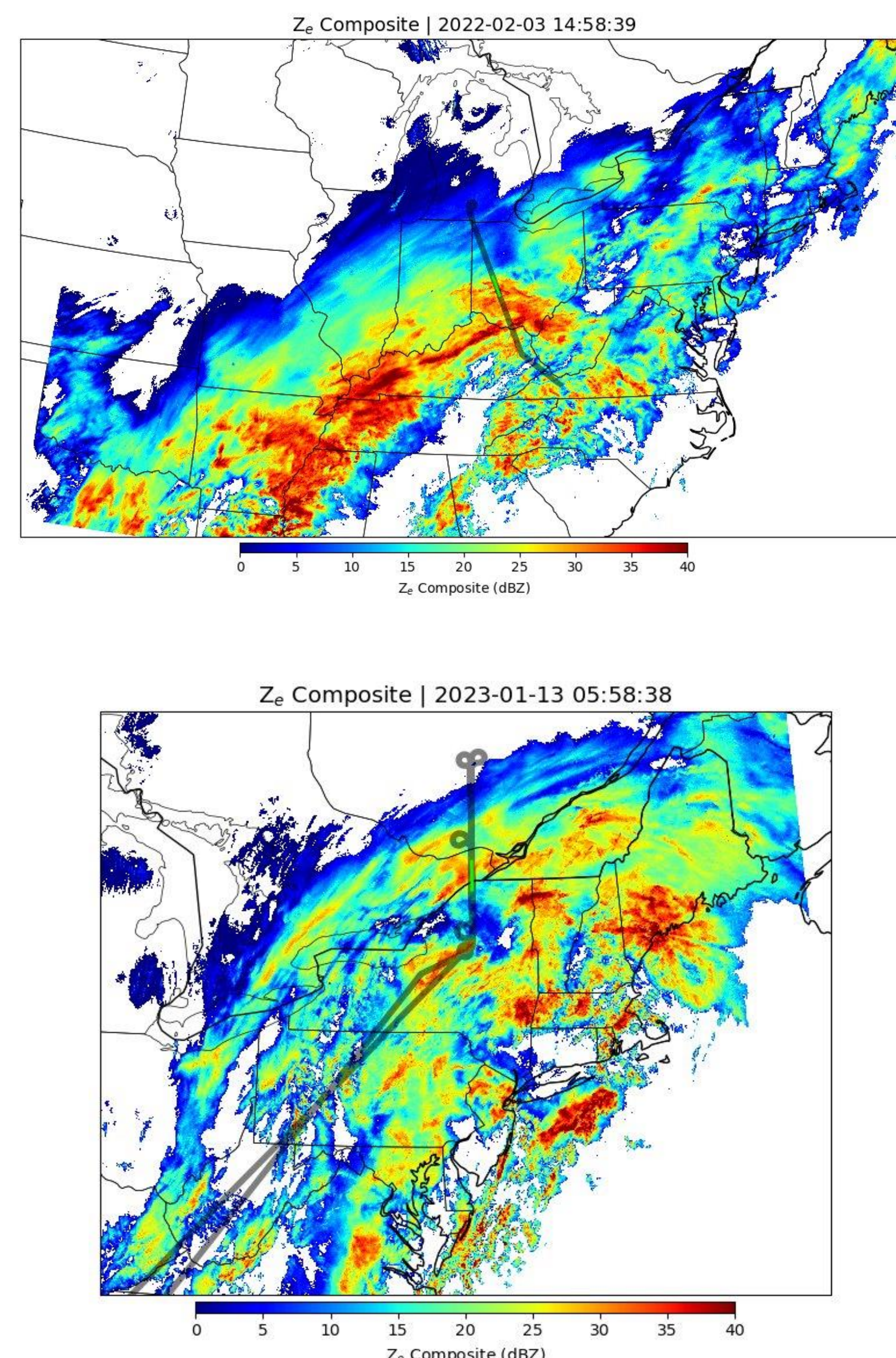


Fig 2 ER2 (grey) flight track on (top) 3 Feb 2022 and (bottom) 13 Jan 2023

3 February 2022

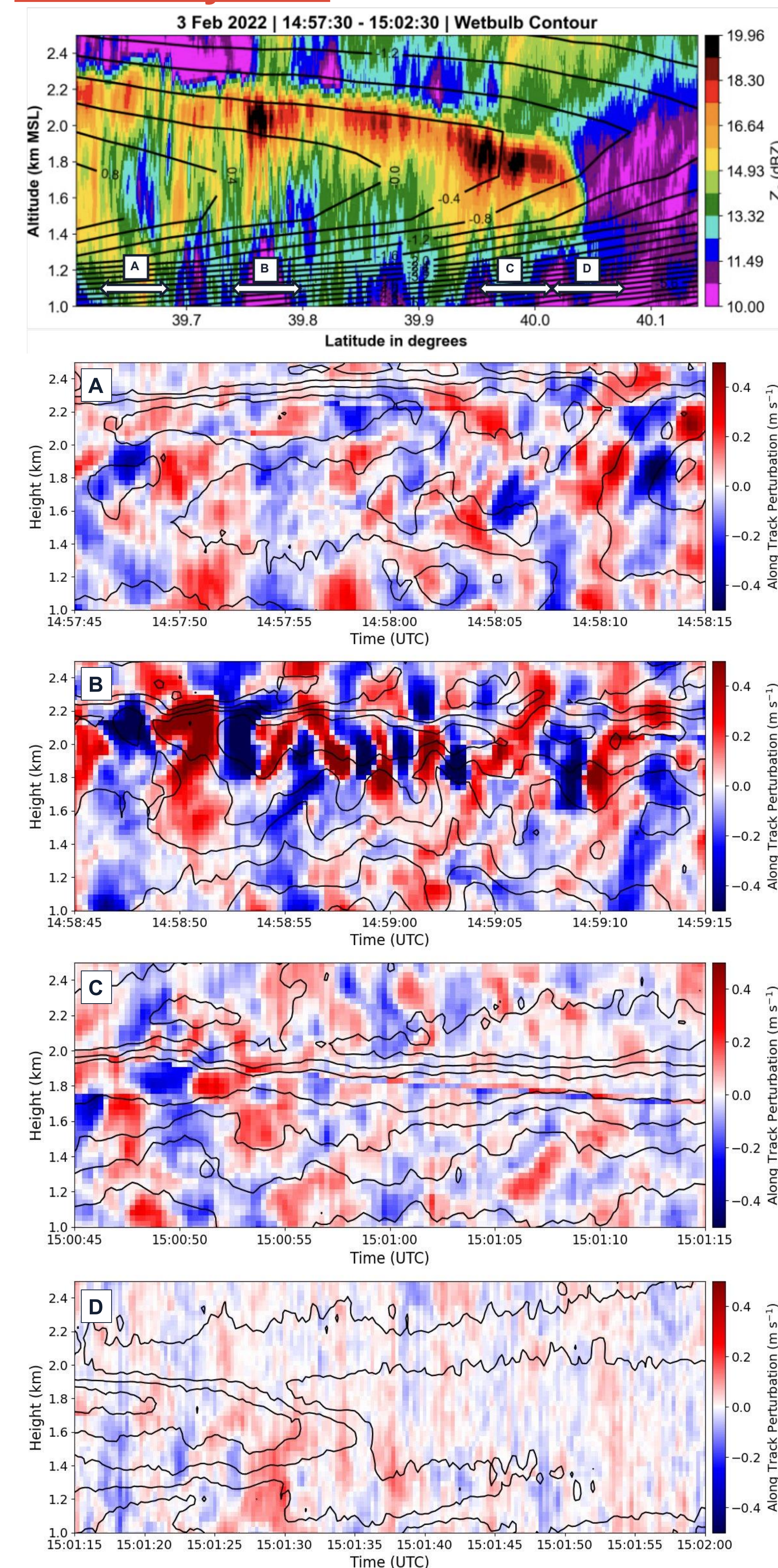


Fig. 3. Reflectivity overlaid with wet-bulb temperature from the HRRR analysis (top panel). Along-track radial velocity perturbations within the melting layer are shown for smaller subsets of the flight track (a–d), with reflectivity overlaid. The corresponding flight track locations are indicated in the top panel.

13 January 2023

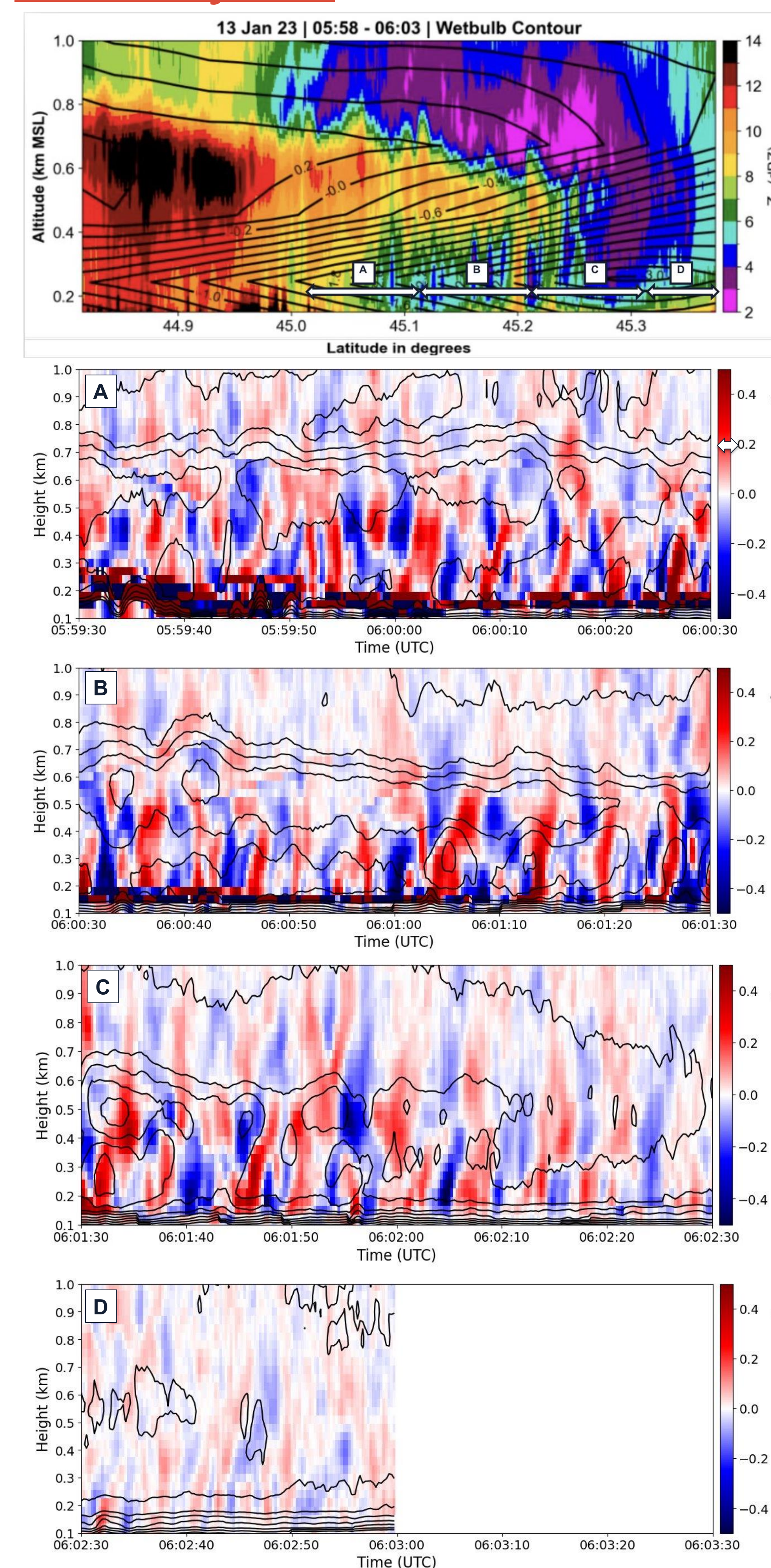


Fig. 4. Same as Figure 3 but for 13 January 2023.

Results

- Plots A through C in both panels show circulations at an approximate 1 to 1 ratio indicating that they occur on a horizontal scale of approximately 100 to 400 m, with perturbations of vertical radial velocity of the order of the 0.1 to 0.5 m/s.
- The perturbations are common across the melting layer and essentially disappear beyond the cold end termination point of the melting layer where only snowfall is occurring.
- These perturbations could be driven by both microphysical and dynamical causes. Microphysical causes include variations in particle melting rate that can lead to variations in particle fall speeds, and/or variations in particle type entering the melting layer. Dynamical causes are driven by latent cooling associated with melting, evaporative cooling below the melting layer, and turbulence due to shear across the melting layer. It is possible that the circulations are associated with gravity waves induced by shear or convective circulations driven by latent cooling.
- Future work includes examining more cases, diagnosing stability near the melting layer, quantifying perturbation structures and magnitudes, and conducting model simulations to isolate the processes leading to the observed fluctuations.

Acknowledgements

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